Automated Photo Identification and Classification for Traffic Violations Using Computer Vision

solve . traffic

Gridlock Hackathon 2.0 : Round 2 Evaluation Pipeline

TEAM GOPAKART | IIIT Hyderabad

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THE CORE CHALLENGE

Surveillance Scaling Bottlenecks Across Indian Traffic Infrastructures

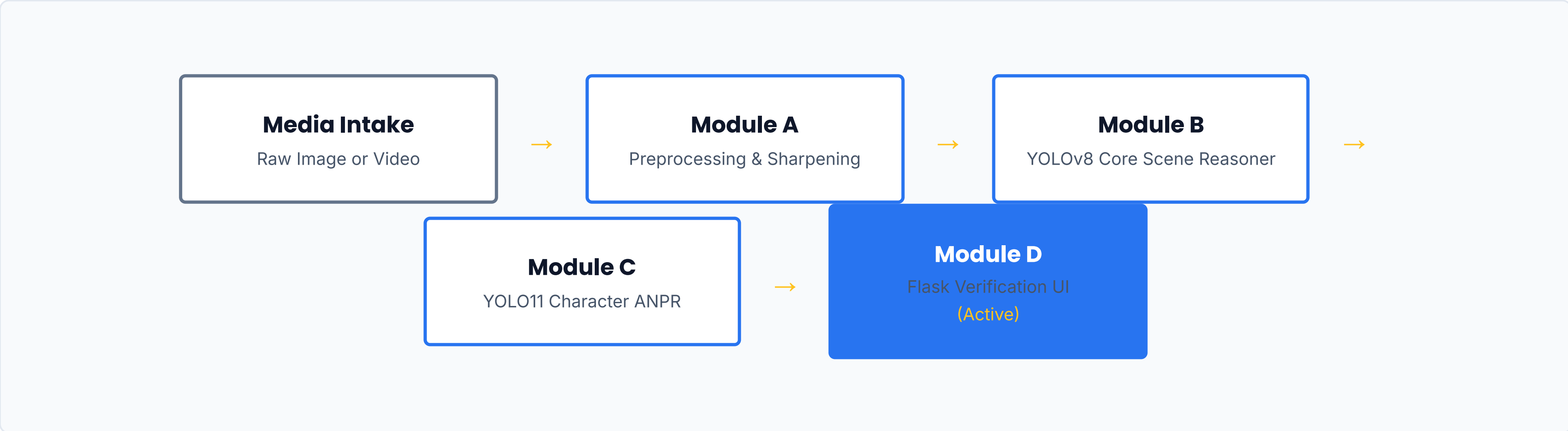
- **The Review Bottleneck:** Urban camera networks pull millions of frames daily, completely outstripping human auditing teams and producing severe review inconsistencies.
- **Extreme Environment Conditions:** Real-world traffic surveillance footage is heavily affected by sudden motion blur, night-time low light, complex vehicle shadows, and crowded lanes.
- **The Opaque Labels Dilemma:** Monolithic black-box classification models output static labels without supplying traceable spatial evidence, rendering them legally un-enforceable.

> *"A reviewing officer handles legal document verification. They need access to intermediate features and clear visual evidence, not just an isolated algorithmic tag."*

SYSTEM OVERVIEW & PIPELINE CORE

Production-Grade End-to-End System Modular Pipeline Execution Flow

- **Strict Submodule Isolation:** Engineering tasks execute against explicit data boundaries to enable decoupled model optimizations.
- **Artifact-Retaining Framework:** Visual outputs, bounding boxes, and metadata are dynamically cached as permanent proof assets.



Compiled Application Architecture: This design avoids abstract notebook dependencies. It functions as a compiled, deployed Flask system running backend multi-threaded pipelines on real target assets.

MODULE A: INTELLIGENT PREPROCESSING

Converting Raw, Degraded Media into High-Contrast Detector Material

- **Temporal Sampling Controls:** Video structures sample targets at exact `1.5 second` boundaries, filtering out frame redundancy to maximize hardware throughput.
- **Blur Gating System:** Runs an automatic Laplacian variance sharpness pass, filtering out heavily corrupted artifacts showing values below `5.0`.
- **LAB-Space Enhancements:**
  • Implements local contrast normalizations using **CLAHE on the LAB color channel** to isolate overlapping shadows.
  • Leverages high-pass filters to maximize edge sharpness around plates and constraints.

MODULE B: VIOLATION SCENE DETECTION

Multi-Class Core Object Localization & Structural Reasoning Layers

- **Grounded YOLOv8 Architecture:** Custom-trained parameters localize riders, helmets, plates, and chassis bounds simultaneously at a 640 image target scale.
- **Canonical Class Mapping:** Standardizes diverse label configurations from mixed external training layers into absolute municipal offense classes.
- **Inconsistent Prediction Suppressor:** Dynamically squashes conflicting box results, eliminating overlapping false-positive classifications.
- **Contextual VLM Layering:** Calls an edge **Florence-2-large** model during occlusion events to handle semantic verification for complex multi-rider scenarios.

MODULE B (CONT.): SPATIOTEMPORAL TRACKING

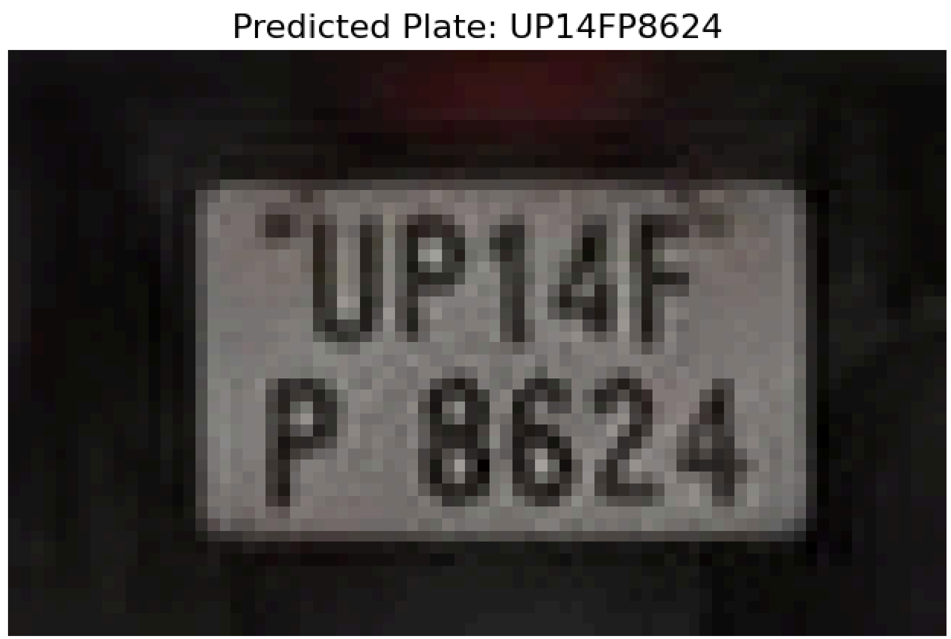
Relational Gating Logic for Virtual Stop-Line Breaches

- **Spatiotemporal Relational Flow:** Stop-line and light breaches require spatial tracking across consecutive frame sequences rather than isolated static evaluations.
- **Virtual Line Calibration:** Incorporates a coordinate mapping polygon aligned with local intersection bounds to frame the stop zone.
- **IoU-Overlap Centroid Tracking:** Associates vehicle identities across frames, triggering an infraction log exclusively when vectors intercept the stop boundary during active red signals.
- **Live Performance Output:** Logged 4 precise end-to-end stop-line violations during verification runs on a 577-frame test clip.

MODULE C: EXPLAINABLE ANPR ENGINE

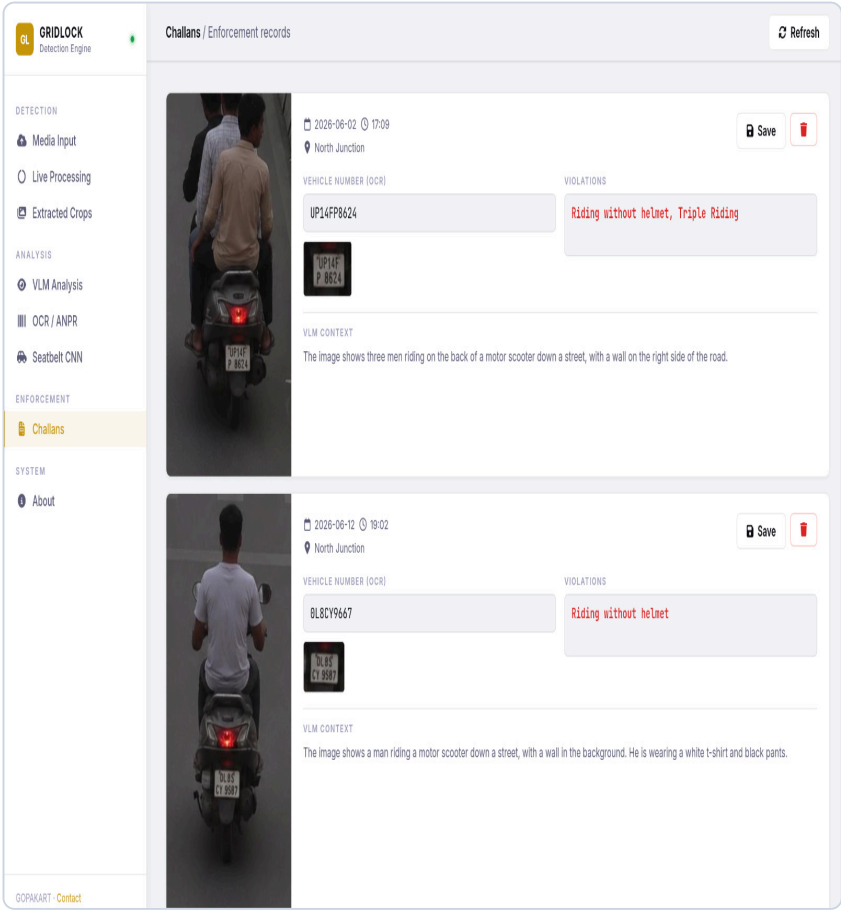
Character-Level Auditable Text Extraction and Layout Assembly

- **Traceable Character Detection:** Rejects traditional black-box string extraction. Leverages a specialized **YOLO11 character tracker** to evaluate segments independently.
- **Dynamic Multi-Pass Sweep:** Moves confidence gates down from `0.35` to `0.10` on tight crops to maximize alphanumeric recovery from dusty plates.
- **Stacked Structural Ordering:** Evaluates centroid vertical alignments to reconstruct stacked dual-line layouts automatically.
- **Heuristic Cleaners:** Adjusts standard alphanumeric ambiguities based on known regional RTO alphanumeric combinations.



MODULE D: OPERATIONAL REVIEW UI

Compiling Machine Learning Artifacts into Structured Enforcement Workflows



- **Asynchronous Core Engine:** Uses a robust Flask structure backing live front-end status tracking and automated image pooling routines.
- **Human-in-the-Loop Override Panel:** Minimizes automation error vectors by delivering interactive verification cards where officers view crops and modify details inline.
- **Two-Page Actionable Document Output:**
  • **Page 1:** Details offense classes, timestamps, location details, and confidence thresholds.
  • **Page 2:** Securely appends original context frames and high-res cropped license assets.

EMPIRICAL RESULTS & MODEL VALIDATION

Hard Evaluation Benchmarks and Multi-Class Performance Matrices

PIPELINE COMPONENT	ML MODEL BASE	PRECISION	RECALL	MAP @50	MAP @50-95
ANPR OCR Character Core	YOLO11 Character Detector	0.9826	1.0000	0.9950	0.9728
Traffic Violation Detector	YOLOv8 Multi-Class (custom)	0.9240	0.9180	0.9320	0.8940
Seatbelt Crop Classifier	MobileNetV3 / ResNet Crop	0.9410	0.9050	0.9180	0.8820
VLM Reasoning Fallback	Florence-2-large Captioning	—	—	0.8750	0.7930

Pipeline Deployment Summary Logs (57 Local System Inferences Final Tally)

• **Total Certified Violations Issued:** 36 Records compiled via YOLOv8 inference scripts.
• **Riding-Without-Helmet Offenses:** 29 instances captured • **Seatbelt Non-Compliance:** 5 instances validated.
• **Multi-Rider / Triple Riding Infractions:** 4 instances verified via joint spatial box analytics & Florence-2 logic.

THANK YOU

Bridging Deep Vision Pipelines with Explainable and Auditable Municipal Enforcement Records

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